

Supplemental Example: Systems of axial bars

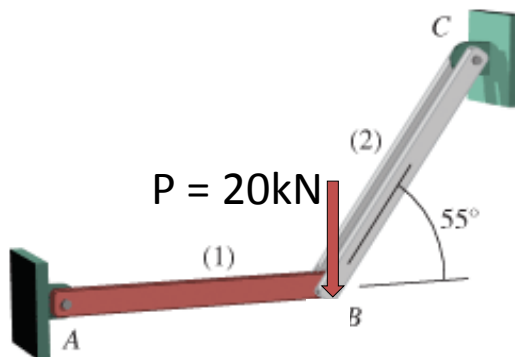
Many of you were in ENME202 last year, while others would have done a similar mechanics of materials course. Here we will look at a simple system of two axial bars that support a downwards load. We will solve the problem using the methods you would have used previously and then re-solve using the finite element method.

Both methods will give the same answer as they rely on the same fundamental underlying theory. However, the implementation of this theory is very different in the two methods.

The intention of this example is to relate the new matrix methods developed in ENME302 to the methods you have previously used (and of course will be very comfortable and familiar with....)

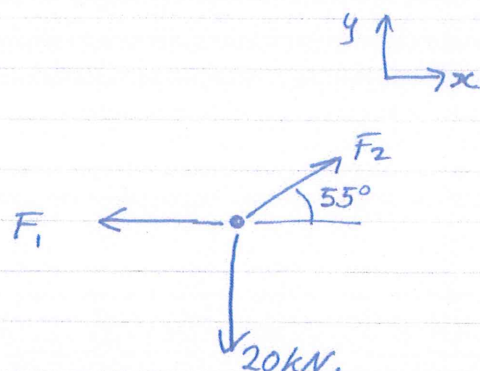
Question 3: The two rods shown are fixed via pin connections. Rod (1) is stainless steel, with length, $L_1 = 1.1$ m, elastic modulus, $E_1 = 200$ GPa, and cross-sectional area, $A_1 = 400$ mm². Rod (2) is mild steel, with $L_2 = 0.8$ m, elastic modulus, $E_2 = 200$ GPa, and cross-sectional area, $A_2 = 600$ mm²

A vertically downwards load of 20 kN is applied at point B, determine the vertical deflection of point B.



Question 3:

FBD of joint B:



$$\uparrow \sum F_y = 0: F_2 \sin 55^\circ - 20000 = 0.$$

$$F_2 = \frac{20000}{\sin 55^\circ}$$

$$\underline{F_2 = 24415 \text{ N.}}$$

$$\rightarrow \sum F_x = 0: F_2 \cos 55^\circ - F_1 = 0.$$

$$F_1 = F_2 \cos 55^\circ$$

$$F_1 = (24415) \cos 55^\circ$$

$$\underline{F_1 = 14,000 \text{ N}}$$

Let us first solve this problem as we would have done in ENME202, for reference.

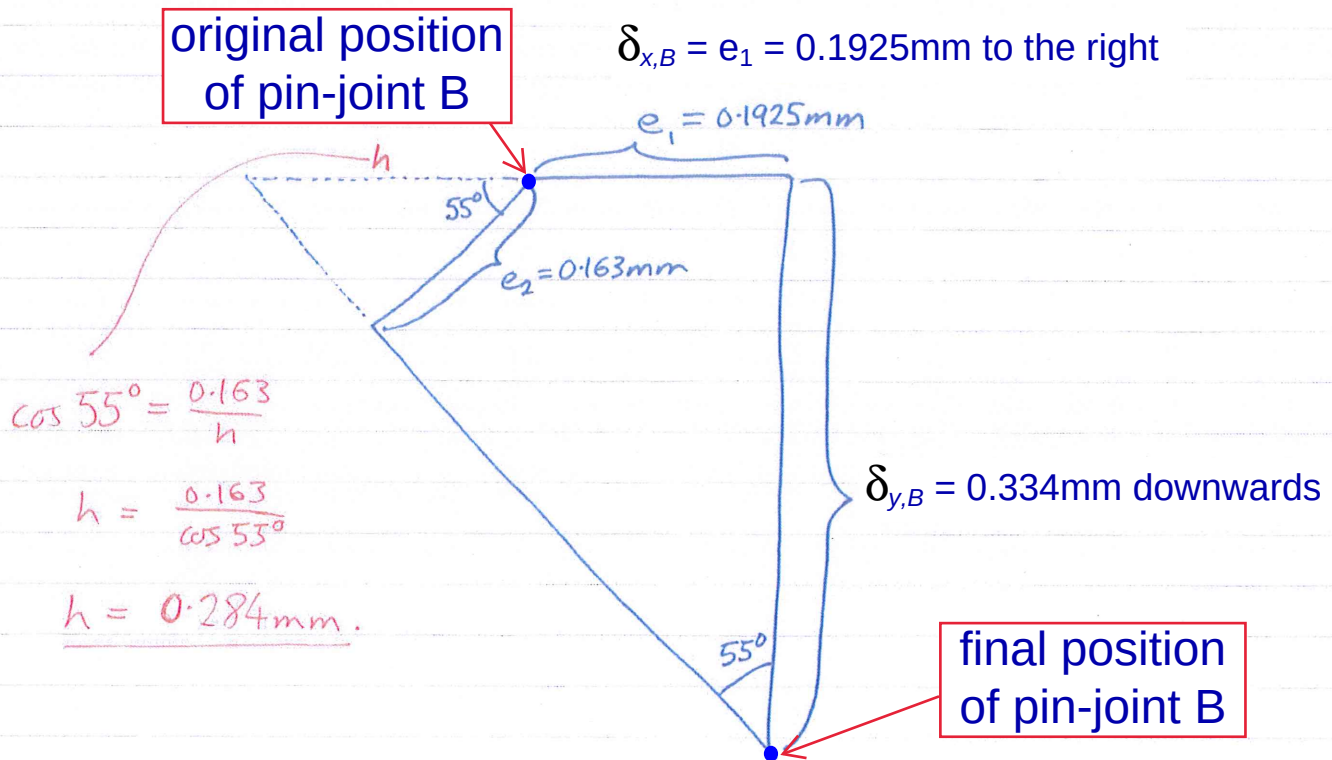
Member elongations: $e_1 = \frac{F_1 L_1}{A_1 E_1} = \frac{(14000)(1.1)}{(400 \times 10^{-6})(200 \times 10^9)}$

$$\underline{e_1 = 0.1925 \text{ mm.}}$$

$$e_2 = \frac{F_2 L_2}{A_2 E_2} = \frac{(24415)(0.8)}{(600 \times 10^{-6})(200 \times 10^9)}$$

$$\underline{e_2 = 0.163 \text{ mm}}$$

Consider the following triangle to determine the vertical deflection at B:



$$\tan 55^\circ = \frac{h + e_1}{\delta_{y,B}}$$

$$\delta_{y,B} = \frac{h + e_1}{\tan 55^\circ}$$

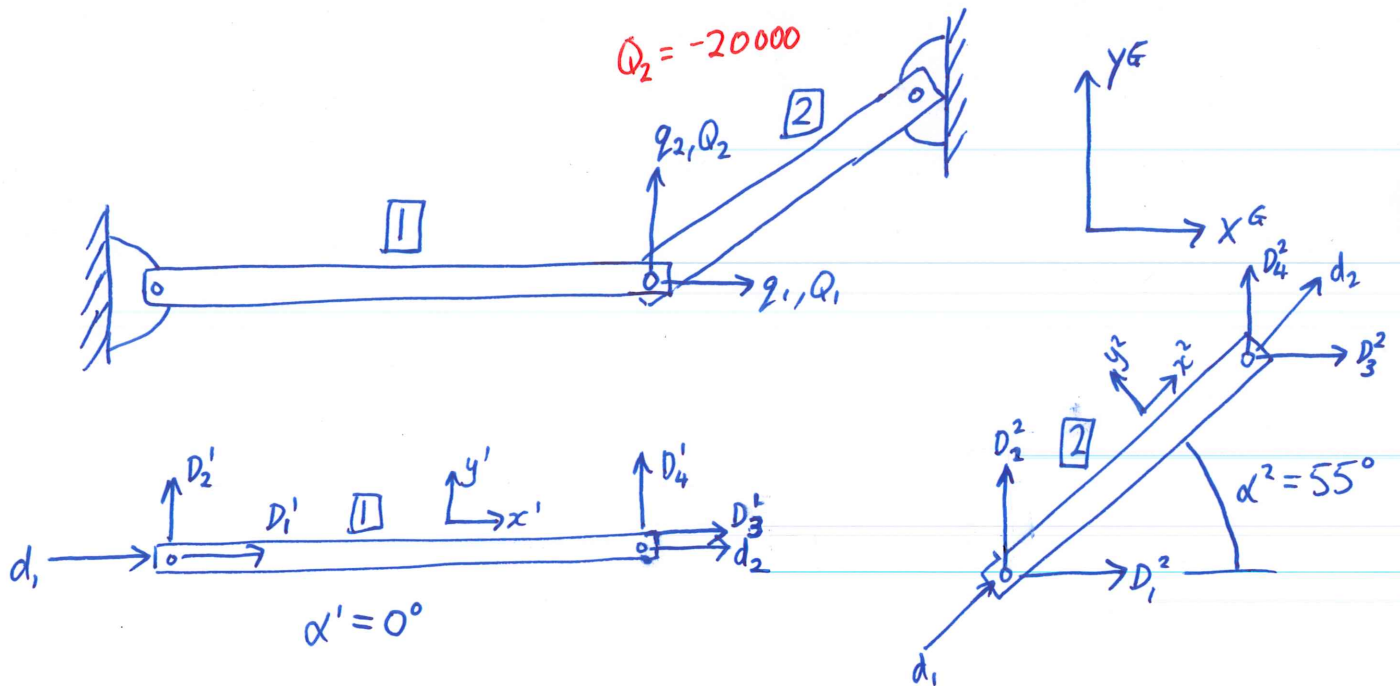
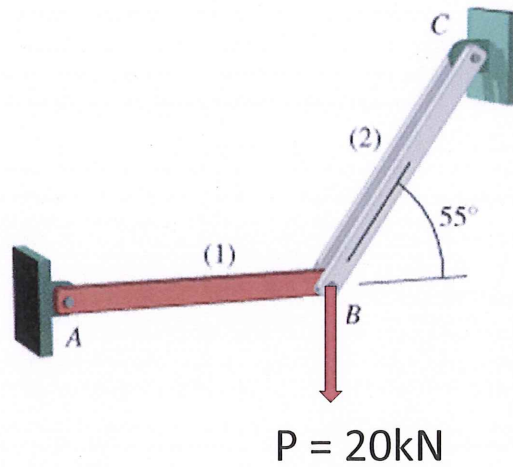
$$\delta_{y,B} = \frac{(0.284) + (0.1925)}{\tan 55^\circ}$$

$$\underline{\delta_{y,B} = 0.334 \text{ mm.}} \text{ vertically downwards}$$

Now we have this result as a reference value, we will solve the problem using finite element methods.

Question 3: The two rods shown are fixed via pin connections. Rod (1) is stainless steel, with length, $L_1 = 1.1$ m, elastic modulus, $E_1 = 200$ GPa, and cross-sectional area, $A_1 = 400$ mm². Rod (2) is mild steel, with $L_2 = 0.8$ m, elastic modulus, $E_2 = 200$ GPa, and cross-sectional area, $A_2 = 600$ mm²

A vertically downwards load of 20kN is applied at point B, determine the vertical deflection of point B.

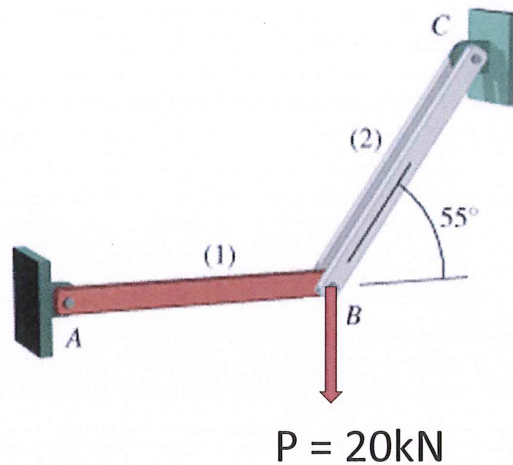


$$A_1 = \begin{bmatrix} \frac{D_1}{L_1} & \frac{D_2}{L_1} & \frac{D_3}{L_1} & \frac{D_4}{L_1} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_2 = \begin{bmatrix} \frac{D_1}{L_2} & \frac{D_2}{L_2} & \frac{D_3}{L_2} & \frac{D_4}{L_2} \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

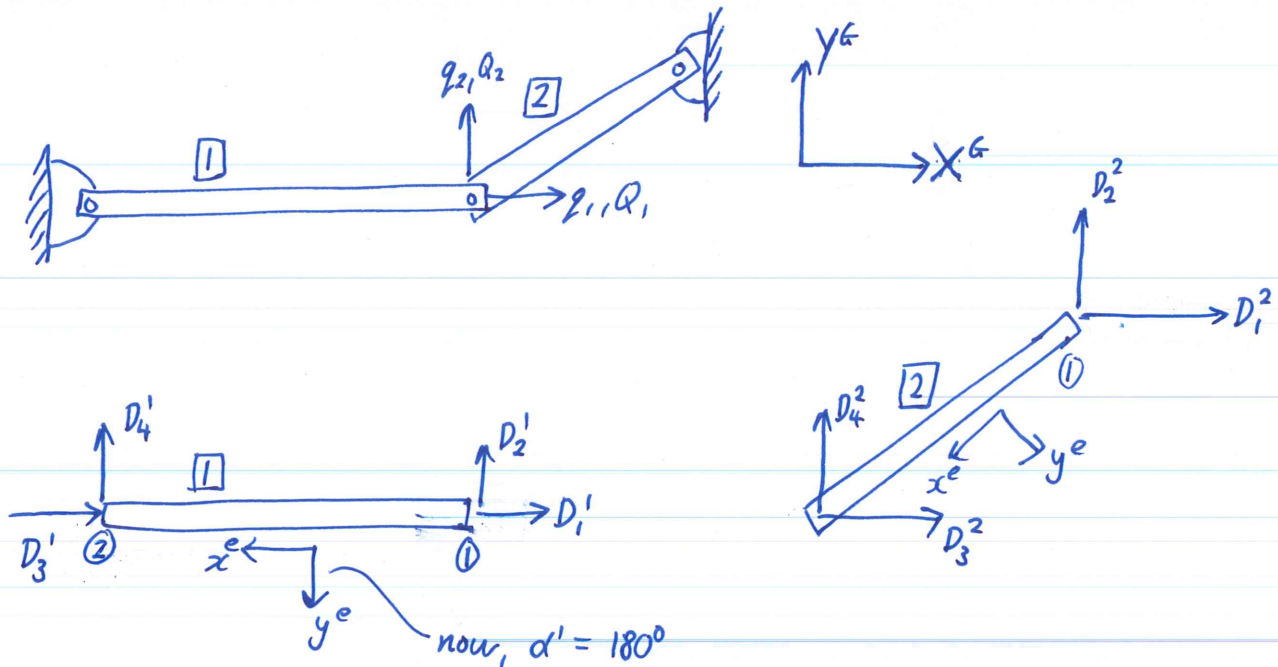
Element 1	Element 2
K1 = Local_BAR(200e9,400e-6,1.1) K1 = 1.0e+07 × 7.2727 -7.2727 -7.2727 7.2727	K2 = Local_BAR(200e9,600e-6,0.8) K2 = 1.0e+08 × 1.5000 -1.5000 -1.5000 1.5000
K1hat = global_bar(K1,alpha1) = global_bar(K1,0) K1hat = 1.0e+07 × 7.2727 0 -7.2727 0 0 0 0 0 -7.2727 0 7.2727 0 0 0 0 0	K2hat = global_bar(K2,alpha2) = global_bar(K2,55) K2hat = 1.0e+08 × 0.4935 0.7048 -0.4935 -0.7048 0.7048 1.0065 -0.7048 -1.0065 -0.4935 -0.7048 0.4935 0.7048 -0.7048 -1.0065 0.7048 1.0065
A1 = np.array([[0, 0, 1, 0],[0, 0, 0, 1]]) (from problem definition) KG1 = A1 @ K1hat @ A1.T or KG1 = A1 @ K1hat @ np.transpose(A1) KG1 = 1.0e+07 × 7.2727 0 0 0	A2 = np.array([[1, 0, 0, 0],[0, 1, 0, 0]]) (from problem definition) KG2 = A2 @ K2hat @ A2.T or KG2 = A2 @ K2hat @ np.transpose(A2) KG2 = 1.0e+08 × 0.4935 0.7048 0.7048 1.0065
KG = KG1 + KG2 = 1.0e+08 × 1.2208 0.7048 0.7048 1.0065	
From problem definition, 0kN applied load in direction of q₁, and 20kN applied load in direction of q₂ $Q = \begin{Bmatrix} 0 \\ -20,000 \end{Bmatrix}$	
q = inv(KG)*Q (use np.linalg.solve to solve equations) q = 1.0e-03 × 0.1926 -0.3335	
F1 = K1hat @ A1.T @ q F1 = 1.0e+04 × -1.4004 0 1.4004 0	F2 = K2hat @ A2.T @ q F2 = 1.0e+04 × -1.4004 -2.0000 1.4004 2.0000
theta1 = 0*np.pi/180, or theta1 = np.radians(0) noting that angle must be in radians Lambda1 = [cos(theta1) sin(theta1) 0 0; 0 0 cos(theta1) sin(theta1)] Lambda1 = 1 0 0 0 0 0 1 0	Theta2 = 55*np.pi/180, or theta2 = np.radians(55) noting that angle must be in radians Lambda2 = [cos(theta2) sin(theta2) 0 0; 0 0 cos(theta2) sin(theta2)] Lambda2 = 0.5736 0.8192 0 0 0 0 0.5736 0.8192
d1 = Lambda1 @ A1.T @ q d1 = 1.0e-03 × 0 0.1926	d2 = Lambda2 @ A2.T @ q d2 = 1.0e-03 × -0.1628 0
f1 = K1 @ d1 f1 = 1.0e+04 × -1.4004 1.4004	f2 = K2 @ d2 f2 = 1.0e+04 × -2.4415 2.4415

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A vertically downwards load of 20kN is applied at point B, determine the vertical deflection of point B.



What if we had chosen a different orientation of the elements??



$$A_1 = \begin{bmatrix} l_1 & 0 \\ l_2 & 0 \end{bmatrix} \begin{bmatrix} D_1' & D_2' & D_3' & D_4' \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} l_1 & 0 \\ l_2 & 0 \end{bmatrix} \begin{bmatrix} D_1^2 & D_2^2 & D_3^2 & D_4^2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

α changes for each element and so does the assembly matrix for each element.

$\Rightarrow$ The final answer will not change!!

Element 1	Element 2
$K1 = \text{Local_BAR}(200\text{e}9, 400\text{e-}6, 1.1)$ $K1 = 1.0\text{e+}07 \times$ $\begin{bmatrix} 7.2727 & -7.2727 \\ -7.2727 & 7.2727 \end{bmatrix}$	$K2 = \text{Local_BAR}(200\text{e}9, 600\text{e-}6, 0.8)$ $K2 = 1.0\text{e+}08 \times$ $\begin{bmatrix} 1.5000 & -1.5000 \\ -1.5000 & 1.5000 \end{bmatrix}$
$K1\text{hat} = \text{global_bar}(K1, \alpha1) = \text{global_bar}(K1, 180)$ $K1\text{hat} = 1.0\text{e+}07 \times$ $\begin{bmatrix} 7.2727 & 0 & -7.2727 & 0 \\ 0 & 0 & 0 & 0 \\ -7.2727 & 0 & 7.2727 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	$K2\text{hat} = \text{global_bar}(K2, \alpha2) = \text{global_bar}(K2, 235)$ $K2\text{hat} = 1.0\text{e+}08 \times$ $\begin{bmatrix} 0.4935 & 0.7048 & -0.4935 & -0.7048 \\ 0.7048 & 1.0065 & -0.7048 & -1.0065 \\ -0.4935 & -0.7048 & 0.4935 & 0.7048 \\ -0.7048 & -1.0065 & 0.7048 & 1.0065 \end{bmatrix}$
$A1 = \text{np.array}([[1, 0, 0, 0], [0, 1, 0, 0]])$ (from problem definition)	$A2 = \text{np.array}([[0, 0, 1, 0], [0, 0, 0, 1]])$ (from problem definition)
$KG1 = A1 @ K1\text{hat} @ A1.T$ or $KG1 = A1 @ K1\text{hat} @ \text{np.transpose}(A1)$ $KG1 = 1.0\text{e+}07 \times$ $\begin{bmatrix} 7.2727 & 0 \\ 0 & 0 \end{bmatrix}$	$KG2 = A2 @ K2\text{hat} @ A2.T$ or $KG2 = A2 @ K2\text{hat} @ \text{np.transpose}(A2)$ $KG2 = 1.0\text{e+}08 \times$ $\begin{bmatrix} 0.4935 & 0.7048 \\ 0.7048 & 1.0065 \end{bmatrix}$
$KG = KG1 + KG2 = 1.0\text{e+}08 \times$ $\begin{bmatrix} 1.2208 & 0.7048 \\ 0.7048 & 1.0065 \end{bmatrix}$	
From problem definition, 0kN applied load in direction of q_1 , and 20kN applied load in direction of q_2 $Q = \begin{Bmatrix} 0 \\ -20,000 \end{Bmatrix}$	
$q = \text{inv}(KG) @ Q$ (use <code>np.linalg.solve</code> to solve equations) $q = 1.0\text{e-}03 \times$ $\begin{bmatrix} 0.1926 \\ -0.3335 \end{bmatrix}$	
$F1 = K1\text{hat} @ A1.T @ q$ $F1 = 1.0\text{e+}04 \times$ $\begin{bmatrix} 1.4004 \\ 0 \\ -1.4004 \\ 0 \end{bmatrix}$	$F2 = K2\text{hat} @ A2.T @ q$ $F2 = 1.0\text{e+}04 \times$ $\begin{bmatrix} 1.4004 \\ 2.0000 \\ -1.4004 \\ -2.0000 \end{bmatrix}$
$\theta1 = 180 \times \text{np.pi} / 180$, or $\theta1 = \text{np.radians}(180)$ noting that angle must be in radians $\text{Lambda}1 = [\cos(\theta1) \sin(\theta1) \ 0 \ 0; \ 0 \ 0 \cos(\theta1) \sin(\theta1)]$ $\text{Lambda}1 =$ $\begin{Bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{Bmatrix}$	$\theta2 = 235 \times \text{np.pi} / 180$, or $\theta2 = \text{np.radians}(235)$ noting that angle must be in radians $\text{Lambda}2 = [\cos(\theta2) \sin(\theta2) \ 0 \ 0; \ 0 \ 0 \cos(\theta2) \sin(\theta2)]$ $\text{Lambda}2 =$ $\begin{Bmatrix} -0.5736 & -0.8192 & 0 & 0 \\ 0 & 0 & -0.5736 & -0.8192 \end{Bmatrix}$
$d1 = \text{Lambda}1 @ A1.T @ q$ $d1 = 1.0\text{e-}03 \times$ $\begin{bmatrix} -0.1926 \\ 0 \end{bmatrix}$	$d2 = \text{Lambda}2 @ A2.T @ q$ $d2 = 1.0\text{e-}03 \times$ $\begin{bmatrix} 0 \\ 0.1628 \end{bmatrix}$
$f1 = K1 @ d1$ $f1 = 1.0\text{e+}04 \times$ $\begin{bmatrix} -1.4004 \\ 1.4004 \end{bmatrix}$ $f^1 = \begin{Bmatrix} -14.0 \text{ kN} \\ 14.0 \text{ kN} \end{Bmatrix}$	$f2 = K2 @ d2$ $f2 = 1.0\text{e+}04 \times$ $\begin{bmatrix} -2.4415 \\ 2.4415 \end{bmatrix}$ $f^2 = \begin{Bmatrix} -24.4 \text{ kN} \\ 24.4 \text{ kN} \end{Bmatrix}$